

SEDAR 72 Gulf of Mexico Gag Grouper

Recommendations for considerations of future research are provided below and do not indicate any particular order of priority.

Recreational Landings and Discards data

- Further develop best practices for correcting for prominent peaks and troughs in the earlier part of the time series where uncertainty is high and catch/discard estimates are driven by few but influential intercept records.
- Develop estimates of uncertainty around Headboat discard estimates.
- Develop estimates of uncertainty around estimates of Recreational landings in weight.

Age and length composition

- Quantify and evaluate appropriate modeling and weighting procedures of length and age compositions to ensure age and length composition inputs are representative of the segment of the population being modeled.

Selectivity and catchability of the commercial fleets

- Further investigate and quantify changes in selectivity/catchability through time to improve fit to the discards and length compositions in recent years.
- Attempt to model changes in fishing behavior caused by the implementation of the IFQ through the use of time-varying selectivity and retention.
- Continue data collection from observer programs or electronic monitoring programs.

Selectivity and retention of the recreational fleets

- Further investigate and quantify changes in selectivity/catchability through time to improve fit to the length compositions across the time series.

Combined Video Index

- Continue the development of the Gulf of Mexico Gag Grouper Combined Video Index. Particularly, exploring ways to account for the survey and habitat effects. One possibility discussed during SEDAR72 would be to develop a separate index for the shallow/inshore and deep/offshore components of the surveys to better represent

and track the various segments of the Gulf of Mexico Gag Grouper population through time.

Landings and Discards

- Explore approaches for assigning uncertainty estimates to commercial landings and revisit estimation of historic landings.
- Further investigate best practices for converting historical recreational landings from numbers to weight.

Recreational CPUE indices

- Additional research is needed to investigate if assumptions are appropriate across fulltime series (e.g., targeting, trip length, effects of various regulations, red snapper).

Natural mortality

- Explore ways to better reflect uncertainty about the mortality at age vector.

Red Tide and other sources of episodic mortality

- Continue to improve the way red tide mortality is modeled inside SS (e.g., extending the blocks-on-M method to older ages, including a vector of biomass loss), explore whether additional historical red tide years should also be modelled, and investigate other potential sources of episodic mortality (e.g. cold snaps) on the Gulf of Mexico Gag Grouper population.